6.4.5 Is LU with Partial Pivoting Stable?

6.4.5 Element growth in LU factorization

fit width

The last unit gives a backward error result regarding LU factorization (and, by extention, LU factorization with pivoting):

$$(A + \Delta A) = \check{L}\check{U} \quad \text{with} \quad |\Delta A| \leq \gamma_n |\check{L}||\check{U}|..$$

The question now is: does this mean that LU factorization with partial pivoting is stable? In other words, is ΔA, which we bounded with $|\Delta A| \leq \gamma_n |\check{L}||\check{U}|$, always small relative to the entries of $|A|$? The following exercise gives some insight:

Homework 6.4.5.1. Apply LU with partial pivoting to

$$A = \begin{pmatrix} 1 & 0 & 1 \\ -1 & 1 & 1 \\ -1 & -1 & 1 \end{pmatrix}.$$

Pivot only when necessary.

▾ Solution

Notice that no pivoting is necessary. Eliminating the entries below the diagonal in the first column yields:

$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 0 & -1 & 2 \end{pmatrix}.$$

Eliminating the entries below the diagonal in the second column again does not require pivoting and yields:

$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 4 \end{pmatrix}.$$

Homework 6.4.5.2. Generalize the insights from the last homework to a $n \times n$ matrix. What is the maximal element growth that is observed?

▾ Solution

Consider

$$A = \begin{pmatrix} 1 & 0 & 0 & \cdots & 0 & 1 \\ -1 & 1 & 0 & \cdots & 0 & 1 \\ -1 & -1 & 1 & \cdots & 0 & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ -1 & -1 & & \cdots & 1 & 1 \\ -1 & -1 & & \cdots & -1 & 1 \end{pmatrix}.$$

Notice that no pivoting is necessary when LU factorization with pivoting is performed.

Eliminating the entries below the diagonal in the first column yields:

$$\begin{pmatrix} 1 & 0 & 0 & \cdots & 0 & 1 \\ 0 & 1 & 0 & \cdots & 0 & 2 \\ 0 & -1 & 1 & \cdots & 0 & 2 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & -1 & & \cdots & 1 & 2 \\ 0 & -1 & & \cdots & -1 & 2 \end{pmatrix}.$$

Eliminating the entries below the diagonal in the second column again does not require pivoting and yields:

$$\begin{pmatrix} 1 & 0 & 0 & \cdots & 0 & 1 \\ 0 & 1 & 0 & \cdots & 0 & 2 \\ 0 & 0 & 1 & \cdots & 0 & 4 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & & \cdots & 1 & 4 \\ 0 & 0 & & \cdots & -1 & 4 \end{pmatrix}.$$

Continuing like this for the remaining columns, eliminating the entries below the diagonal leaves us with the upper triangular matrix

$$\begin{pmatrix} 1 & 0 & 0 & \cdots & 0 & 1 \\ 0 & 1 & 0 & \cdots & 0 & 2 \\ 0 & 0 & 1 & \cdots & 0 & 4 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & & \cdots & 1 & 2^{n-2} \\ 0 & 0 & & \cdots & 0 & 2^{n-1} \end{pmatrix}.$$

From these exercises we conclude that even LU factorization with partial pivoting can yield large (exponential) element growth in U.

In practice, this does not seem to happen and LU factorization is considered to be stable.